Week 4

Lecture 9 The kernel and the range

*This covers part of **Section 2.2** and a bit of **Section 2.5** in the course notes.*

Here we study the two fundamental vector spaces associated to a linear map: the kernel and the range. Once we define these, we will establish an important relation between these spaces and the behavior of linear maps as functions on sets.

As usual, fix vector spaces V and W over $\mathbb{F}$.

Definition 9.1 (Kernel and range)

Let $T : V \to W$ be a linear map between two vector spaces over $\mathbb{F}$. The **kernel** (or **null space**) of T is the set:

$$\mathrm{Ker}(T) = \{\vec{v} \in V : T\vec{v} = \vec{0}_W\} \subseteq V$$

and the **range** (or **image**) of T is the set:

$$\mathrm{Range}(T) = \{T\vec{v} : \vec{v} \in V\} \subseteq W$$

Proposition 9.2 (Kernel and range are subspaces)

For $T : V \to W$ a linear map, $\mathrm{Ker}(T)$ is a subspace of V and $\mathrm{Range}(T)$ is a subspace of W.

Proof. Notice that $\vec{0}_V \in \mathrm{Ker}(T)$ since $T\vec{0}_V = \vec{0}_W$ and consequently $\vec{0}_W \in \mathrm{Range}(T)$. So these are both non-empty.

Now for $\vec{v}_1, \vec{v}_2 \in \mathrm{Ker}(T)$ and $\alpha \in \mathbb{F}$, we see that:

$$T(\alpha\vec{v}_1 + \vec{v}_2) = \alpha T\vec{v}_1 + T\vec{v}_2 = \vec{0}_W$$

Hence, $\alpha\vec{v}_1 + \vec{v}_2 \in \mathrm{Ker}(T)$, so $\mathrm{Ker}(T)$ is a subspace of V.

Likewise for $\vec{w}_1, \vec{w}_2 \in \mathrm{Range}(T)$ and $\alpha \in \mathbb{F}$, we have $\vec{w}_1 = T\vec{v}_1$ and $\vec{w}_2 = T\vec{v}_2$ for some $\vec{v}_1, \vec{v}_2 \in V$. So then:

$$\alpha\vec{w}_1 + \vec{w}_2 = \alpha T\vec{v}_1 + T\vec{v}_2 = T(\alpha\vec{v}_1 + \vec{v}_2)$$

which shows $\alpha\vec{w}_1 + \vec{w}_2 \in \mathrm{Range}(T)$ and so $\mathrm{Range}(T)$ is a subspace of W. □

We shall revisit some of the maps we saw earlier and determine their kernels and ranges.

Example 9.3

Consider general V and W.

(1) For the identity map $\mathrm{id}_V : V \to V$, we see that if $\mathrm{id}_V(\vec{v}) = \vec{0}_V$, then $\vec{v} = \vec{0}_V$, so $\mathrm{Ker}(\mathrm{id}_V) = \{\vec{0}_V\}$. Clearly $\mathrm{Range}(\mathrm{id}_V) = V$.

(2) For the zero map $O : V \to W$, it is easy to see that $\mathrm{Ker}(O) = V$ and $\mathrm{Range}(O) = \{\vec{0}_W\}$.

(3) Let U be a subspace of V and consider the inclusion map $\iota : U \to V$ given by $\iota(\vec{u}) = \vec{u}$ for all $\vec{u} \in U$. Then $\mathrm{Ker}(\iota) = \{\vec{0}_U\} = \{\vec{0}_V\}$ and $\mathrm{Range}(\iota) = U$.

Example 9.4

We revisit the map $T : \mathbb{R}^3 \to \mathbb{R}^2$ given by:

$$T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

To find the kernel, suppose:

$$T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{0}.$$

This means:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and thus, $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathrm{Ker}(T)$ if and only if $x = y = 0$. So we have:

$$\mathrm{Ker}(T) = \left\{ \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} : z \in \mathbb{R} \right\} = \mathrm{Span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Similarly it is clear that $\mathrm{Range}(T) = \mathbb{R}^2$, since for any $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$ we have for instance:

$$\begin{bmatrix} x \\ y \end{bmatrix} = T\begin{bmatrix} x \\ y \\ 0 \end{bmatrix}.$$

Example 9.5

Consider the differentiation map $D : \mathcal{P}(\mathbb{F}) \to \mathcal{P}(\mathbb{F})$. Notice that if $D(p(x)) = 0$ then $p'(x) = 0$. This must mean $p(x)$ is constant, so $\text{Ker}(D) = \mathcal{P}_0(\mathbb{F}) = \mathbb{F}$.

For any $p(x) \in \mathcal{P}(\mathbb{F})$, write:

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

and let:

$$q(x) = a_0 x + \frac{a_1}{2} x^2 + \frac{a_2}{3} x^3 + \cdots + \frac{a_n}{n+1} x^{n+1}$$

(notice the above is just $V(p(x))$ where V is the integration operator). Then it is clear that $D(q(x)) = p(x)$. This shows $\text{Range}(D) = \mathcal{P}(\mathbb{F})$.

This example also illustrates why the choice of domain is important. Suppose we restricted D to $\mathcal{P}_n(\mathbb{F})$ $(n \geq 1)$ so that we have the map $D : \mathcal{P}_n(\mathbb{F}) \to \mathcal{P}_n(\mathbb{F})$. Then $\text{Ker}(D) = \mathbb{F}$ again, but now $\text{Range}(D) = \mathcal{P}_{n-1}(\mathbb{F})$. Hence, when we restrict D from being an operator on $\mathcal{P}(\mathbb{F})$ to an operator on $\mathcal{P}_n(\mathbb{F})$, the range is *not* the entire codomain $\mathcal{P}_n(\mathbb{F})$.

Exercise 9.6

Compute the kernel and range of the integration map $V : \mathcal{P}(\mathbb{F}) \to \mathcal{P}(\mathbb{F})$ given by:

$$V(p(x)) = \int_0^x p(t)\, dt$$

and use your findings to verify that the set:

$$\{p(x) \in \mathcal{P}(\mathbb{F}) : p(0) = 0\}$$

is a subspace of $\mathcal{P}(\mathbb{F})$.

We now turn to another way in which we can view the kernel and the range of a linear map. Recall first the following definition for functions on sets.

Definition 9.7 (Injective and surjective functions)

If X and Y are two sets and $f : X \to Y$ is a function, we say f is:

- **injective** (or **one-to-one**) when for every $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$; and
- **surjective** (or **onto**) when for every $y \in Y$, there exists an $x \in X$ such that $f(x) = y$.

In other words, f being injective means any two distinct inputs of f must also have distinct outputs, and f being surjective means every element of the codomain Y is output by f.

Example 9.8

(1) An example of an injective function is the map $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = e^x$. If $f(x_1) = f(x_2)$ then $e^{x_1} = e^{x_2}$, and applying ln to each side gives $x_1 = x_2$.

In this case, f is *not* surjective, since for instance there is no $x \in \mathbb{R}$ giving $f(x) = -1$.

(2) An example of a surjective function is the map $g : \mathbb{Z} \to \mathbb{N}$ given by $g(n) = |n|$, the absolute value of n (here we are assuming $\mathbb{N} = \{0, 1, 2, ...\}$). Every natural number $k \in \mathbb{N}$ can be reached by k itself, as $g(k) = |k| = k$.

On the other hand, g is *not* injective, since for instance $g(1) = |1| = 1$ and $g(-1) = |-1| = 1$ as well.

As it turns out, the kernel and range of a linear map are strongly related to the injectivity and surjectivity of the map.

Proposition 9.9

Let $T : V \to W$ be a linear map. Then:

(i) T is injective if and only if $\text{Ker}(T) = \{\vec{0}_V\}$.

(ii) T is surjective if and only if $\text{Range}(T) = W$.

Proof. For **(i)**, suppose first that T is injective. Let $\vec{v} \in \text{Ker}(T)$. Then:

$$T\vec{v} = \vec{0}_W = T\vec{0}_V$$

so by injectivity, we must have $\vec{v} = \vec{0}_V$. Thus, we have proved that $\text{Ker}(T) = \{\vec{0}_V\}$. Conversely, suppose $\text{Ker}(T) = \{\vec{0}_V\}$. Let $\vec{v}_1, \vec{v}_2 \in V$ and suppose $T\vec{v}_1 = T\vec{v}_2$. Then:

$$\vec{0}_W = T\vec{v}_1 - T\vec{v}_2 = T(\vec{v}_1 - \vec{v}_2)$$

and thus $\vec{v}_1 - \vec{v}_2 \in \text{Ker}(T)$. This forces $\vec{v}_1 - \vec{v}_2 = \vec{0}_V$, so $\vec{v}_1 = \vec{v}_2$ and T is injective. Statement **(ii)** is clear from the definition of $\text{Range}(T)$. □

Remark 9.10

In light of the preceding proposition, if $T : V \to W$ is linear, we can interpret $\text{Ker}(T)$ and $\text{Range}(T)$ as follows:

- $\text{Ker}(T)$ is a measure of "how far away" T is from being *injective*.

  On one extreme, if $\text{Ker}(T) = V$, then we just have $T = O$ (the zero map) which is the *least injective* a map can be (there is *no* distinct $\vec{v}_1, \vec{v}_2 \in V$ giving $T\vec{v}_1 \neq T\vec{v}_2$ in this case).

  On the other extreme, $\text{Ker}(T) = \{\vec{0}_V\}$ means precisely that T is injective.

Takeaway: the closer $\text{Ker}(T)$ is to being $\{\vec{0}_V\}$, the "more injective" T is.

- Dually, $\text{Range}(T)$ is a measure of "how far away" T is from being *surjective*.

  On one extreme, if $\text{Range}(T) = W$, then we have surjectivity.

  On the other extreme, if $\text{Range}(T) = \{\vec{0}_W\}$, then we again get that $T = O$ (the zero map), which is the *least surjective* a map can be.

  Takeaway: the closer $\text{Range}(T)$ is to being W, the "more surjective" T is.

Example 9.11

We consider two classic examples of injective and surjective linear maps.

(1) We can often *inject* a vector space into a larger one (more precisely, a space of larger dimension). For example, if we want to inject a copy of $\mathbb{R}^2$ into $\mathbb{R}^4$, we can define $T : \mathbb{R}^2 \to \mathbb{R}^4$ by:

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \\ 0 \end{bmatrix}$$

It is easy to check T is linear and we see that $\text{Ker}(T) = \{\vec{0}\}$ since the only way to obtain $T\vec{x} = \vec{0}$ is for both the x and y coordinates of $\vec{x}$ to be 0. Hence, T is injective.

Notice there are several ways to inject. We chose to inject onto the first two coordinates here, but we can also take something like:

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ 0 \\ x \\ 0 \end{bmatrix}$$

(2) We can also go the other way and *project* a larger space onto a smaller one. For instance, $T : \mathbb{R}^4 \to \mathbb{R}^2$ given by:

$$T\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} w \\ x \end{bmatrix}$$

One can check this is also linear (in fact, we have seen a version of this earlier where we projected $\mathbb{R}^3$ onto $\mathbb{R}^2$). Moreover, this is clearly surjective.

Similarly, this is not the only way we can project. We can also do for example:

$$T\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} z \\ y \end{bmatrix}$$

Example 9.12

Consider the differentiation map $D : \mathcal{P}(\mathbb{F}) \to \mathcal{P}(\mathbb{F})$. We saw that $\text{Ker}(D) = \mathcal{P}_0(\mathbb{F}) = \mathbb{F}$. This subspace can be thought of as a "line" – a one-dimensional subspace spanned by the vector 1.

Now consider the double differentiation map $D^2 = D \circ D : \mathcal{P}(\mathbb{F}) \to \mathcal{P}(\mathbb{F})$ given by $D^2(p(x)) = p''(x)$ (the second derivative of $p(x)$). We can check that $\text{Ker}(D^2) = \mathcal{P}_1(\mathbb{F})$ (the space of linear polynomials), and this is spanned by $\{1, x\}$.

Since $\text{Ker}(D) \subsetneq \text{Ker}(D^2)$, we see that, although neither D nor D^2 are injective, D is "more injective" than D^2.

Exercise 9.13

Determine which of the following linear maps are injective and/or surjective:

(a) $T : \mathbb{F}^3 \to \mathcal{P}_2(\mathbb{F})$ given by:

$$T\begin{bmatrix} a \\ b \\ c \end{bmatrix} = (a+b) + (b+c)x + (a+c)x^2.$$

(b) $S : \mathcal{C}([0,1]) \to \mathbb{R}$ given by:

$$S(f) = \int_0^1 f(t)\, dt.$$

(c) $\tau : \mathcal{M}_n(\mathbb{F}) \to \mathcal{M}_n(\mathbb{F})$ given by:

$$\tau(A) = A + A^T$$

where A^T is the transpose of A.

Exercise 9.14

Recall that:

$$\mathcal{C}^1([0,1]) = \{f \in \mathcal{C}([0,1]) : f \text{ is differentiable and } f' \text{ is continuous}\}.$$

Consider the map $T : \mathcal{C}^1([0,1]) \to \mathcal{C}([0,1])$ given by:

$$T(f) = f + f'$$

for every $f \in \mathcal{C}^1([0,1])$.

(a) Show that T is linear by expressing it as a sum of two linear functions.

(b) Find $\mathrm{Ker}(T)$ and show that this has dimension 1.

(c) Is T surjective? Prove or disprove.

Lecture 10 The rank-nullity theorem

This covers part of ***Section 2.2*** *in the course notes.*

Fix V and W vector spaces over $\mathbb{F}$. We continue studying the kernel and range of a linear map $T : V \to W$, but when one (or both) of V or W is finite-dimensional.

For the situation where V is finite-dimensional, the kernel and range are sure to be finite-dimensional as well.

Proposition 10.1

Let $T : V \to W$ be a linear map and suppose V is finite-dimensional with $\dim(V) = n$. Then $\text{Ker}(T)$ and $\text{Range}(T)$ are also finite-dimensional. In particular, $\text{Ker}(T)$ and $\text{Range}(T)$ each have dimension $\leq n$.

Proof. That $\text{Ker}(T)$ is finite-dimensional is clear from the fact that any subspace of a finite-dimensional subspace is also finite-dimensional with smaller or equal dimension, so $\dim(\text{Ker}(T)) \leq n$.

To see that $\text{Range}(T)$ is finite-dimensional, let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for V and let $\vec{w}_i = T\vec{v}_i \in \text{Range}(T)$ for each $1 \leq i \leq n$. We claim that $\{\vec{w}_1, ..., \vec{w}_n\}$ is a spanning set for W. Indeed, given any $\vec{w} \in \text{Range}(T)$, we have $\vec{w} = T\vec{v}$ for some $\vec{v} \in V$. Now we can write:

$$\vec{v} = \alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n$$

for some $\alpha_i \in \mathbb{F}$, and consequently:

$$\begin{aligned}\vec{w} &= T\vec{v} \\ &= T(\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n) \\ &= \alpha_1 T\vec{v}_1 + \cdots + \alpha_n T\vec{v}_n \\ &= \alpha_1\vec{w}_1 + \cdots + \alpha_n\vec{w}_n \in \text{Span}(\{\vec{w}_1, ..., \vec{w}_n\})\end{aligned}$$

This also shows $\dim(\text{Range}(T)) \leq n$ because we know any spanning set contains a basis (of smaller or equal size). □

Definition 10.2 (Rank and nullity)

Let $T : V \to W$ be a linear map. We define the **rank** of T to be $\text{rank}(T) = \dim(\text{Range}(T))$ and the **nullity** of T to be $\text{nullity}(T) = \dim(\text{Ker}(T))$.

If V is finite-dimensional, then by **Proposition 10.1**, we can be sure that $\text{rank}(T)$ and $\text{nullity}(T)$ are finite values and that they are each $\leq \dim(V)$.

Example 10.3

(1) We saw that for $n \geq 1$, the differentiation map $D : \mathcal{P}_n(\mathbb{F}) \to \mathcal{P}_n(\mathbb{F})$ has $\text{Ker}(D) = \mathbb{F}$ and $\text{Range}(D) = \mathcal{P}_{n-1}(\mathbb{F})$. Therefore, $\text{rank}(D) = n$ and $\text{nullity}(D) = 1$.

(2) The map $T : \mathbb{R}^3 \to \mathbb{R}^2$ given by:

$$T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

we saw earlier had $\text{Range}(T) = \mathbb{R}^2$ and:

$$\text{Ker}(T) = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Thus, $\text{rank}(T) = 2$ and $\text{nullity}(T) = 1$.

We have a special relation between $\text{rank}(T)$ and $\text{nullity}(T)$ In each of the examples given above, notice that the sum of the rank and nullity gives us precisely the dimension of the domain. It turns out this is not just a coincidence – this is always true!

Theorem 10.4 (Rank-nullity theorem)

Let $T : V \to W$ be a linear map and suppose V is finite-dimensional with $\dim(V) = n$. Then $\text{rank}(T) + \text{nullity}(T) = n$.

Proof. We know that $\text{rank}(T), \text{nullity}(T) \leq n$. So fix a basis $\mathcal{B} = \{\vec{u}_1, ..., \vec{u}_k\}$ for $\text{Ker}(T)$ where $k \leq n$. We wish to prove that $\text{rank}(T) = n - k$.

Since $\mathcal{B}$ is a linearly-independent set in V, we can extend it to a basis:

$$\widetilde{\mathcal{B}} = \{\vec{u}_1, ..., \vec{u}_k, \vec{v}_1, ..., \vec{v}_{n-k}\}$$

for V. For each $1 \leq j \leq n - k$, let $\vec{w}_j = T\vec{v}_j$ and set $\mathcal{C} = \{\vec{w}_1, ..., \vec{w}_{n-k}\}$. It is clear that $\mathcal{C}$ is a subset of $\text{Range}(T)$. We claim that in fact $\mathcal{C}$ is a basis for $\text{Range}(T)$. We have two steps to establish this:

Step 1: show that $\mathcal{C}$ is a spanning set for $\text{Range}(T)$.

Let $\vec{w} \in \text{Range}(T)$. Then $\vec{w} = T\vec{v}$ for some $\vec{v} \in V$. Since $\widetilde{\mathcal{B}}$ is a basis for V, we can write:

$$\vec{v} = \alpha_1\vec{u}_1 + \cdots + \alpha_k\vec{u}_k + \beta_1\vec{v}_1 + \cdots + \beta_{n-k}\vec{v}_{n-k}$$

for coefficients $\alpha_i, \beta_j \in \mathbb{F}$ for $1 \leq i \leq k$ and $1 \leq j \leq n - k$. So then:

$$\begin{aligned} \vec{w} &= T\vec{v} \\ &= T(\alpha_1\vec{u}_1 + \cdots + \alpha_k\vec{u}_k + \beta_1\vec{v}_1 + \cdots + \beta_{n-k}\vec{v}_{n-k}) \\ &= \alpha_1 T\vec{u}_1 + \cdots + T\alpha_k\vec{u}_k + \beta_1 T\vec{v}_1 + \cdots + \beta_{n-k}T\vec{v}_{n-k} \end{aligned}$$

Since each $\vec{u}_i$ is in $\text{Ker}(T)$, all the $T\vec{u}_i$ terms disappear. Also each $T\vec{v}_j$ becomes $\vec{w}_j$ by definition of $\vec{w}_j$. So altogether we get:

$$\vec{w} = \beta_1\vec{w}_1 + \cdots + \beta_{n-k}\vec{w}_{n-k}$$

which makes $\mathcal{C}$ a spanning set.

Step 2: show that $\mathcal{C}$ is independent.

Suppose we have coefficients $\beta_1, ..., \beta_{n-k} \in \mathbb{F}$ satisfying the relation:

$$\beta_1\vec{w}_1 + \cdots + \beta_{n-k}\vec{w}_{n-k} = \vec{0}_W$$

Since $\vec{w}_i = T\vec{v}_i$, the above can be rewritten as:

$$T(\beta_1\vec{v}_1 + \cdots + \beta_{n-k}\vec{v}_{n-k}) = \vec{0}_W$$

This means $\beta_1\vec{v}_1 + \cdots + \beta_{n-k}\vec{v}_{n-k} \in \ker(T)$. So we can find coefficients $\alpha_1, ..., \alpha_k \in \mathbb{F}$ for which:

$$\beta_1\vec{v}_1 + \cdots + \beta_{n-k}\vec{v}_{n-k} = \alpha_1\vec{u}_1 + \cdots + \alpha_k\vec{u}_k$$

Moving the LHS to to the right gives us:

$$\alpha_1\vec{u}_1 + \cdots + \alpha_k\vec{u}_k - \beta_1\vec{v}_1 - \cdots - \beta_{n-k}\vec{v}_{n-k} = \vec{0}_V$$

and independence of $\widetilde{\mathcal{B}}$ forces each α_i and β_i to be 0. Hence, we get independence of $\mathcal{C}$.

So now $\mathcal{C}$ is a basis for $\text{Range}(T)$ and contains $n-k$ elements, and thus we conclude $\text{rank}(T) = n - k$, and the proof is complete. □

Example 10.5

Recall that we found an explicit basis for the subspace

$$\{A \in \mathcal{M}_2(\mathbb{F}) : \text{Tr}(A) = 0\}$$

of $\mathcal{M}_2(\mathbb{F})$ in order to prove that its dimension is 3. Similarly, proving that the subspace:

$$\{A \in \mathcal{M}_3(\mathbb{F}) : \text{Tr}(A) = 0\}$$

has dimension 8 would involve finding a basis for it.

With rank-nullity at our disposal, proving this is much easier. For general $n \geq 1$, we know that $\text{Tr} : \mathcal{M}_n(\mathbb{F}) \to \mathbb{F}$ is a linear map and that

$$W = \{A \in \mathcal{M}_n(\mathbb{F}) : \text{Tr}(A) = 0\} = \text{Ker}(\text{Tr})$$

is indeed a subspace.

To find $\dim(W)$, notice that $\text{Range}(\text{Tr}) = \mathbb{F}$ (any value $a \in \mathbb{F}$ will have that for example $\text{Tr}(aE_{1,1}) = a$). Hence, $\text{rank}(\text{Tr}) = 1$. Also $\dim(\mathcal{M}_n(\mathbb{F})) = n^2$.

So by the rank-nullity theorem, we have $\dim(W) = \text{nullity}(\text{Tr}) = n^2 - 1 = (n+1)(n-1)$.

Example 10.6

Consider $\mathcal{P}_4(\mathbb{R})$ and let:

$$W = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : p(0) = p'(0) = p''(0) = 0\}$$

where $p'(x)$ and $p''(x)$ denotes the first and second derivatives of $p(x)$, respectively. We wish to show that W is a subspace of $\mathcal{P}_4(\mathbb{R})$ and find its dimension.

To do so, we can exhibit W as the kernel of some linear map. Consider $T : \mathcal{P}_4(\mathbb{R}) \to \mathbb{R}^3$ given by:

$$T(p(x)) = \begin{bmatrix} p(0) \\ p'(0) \\ p''(0) \end{bmatrix}.$$

Notice:

$$T(\alpha p(x)+q(x)) = \begin{bmatrix} (\alpha p+q)(0) \\ (\alpha p+q)'(0) \\ (\alpha p+q)''(0) \end{bmatrix} = \alpha \begin{bmatrix} p(0) \\ p'(0) \\ p''(0) \end{bmatrix} + \begin{bmatrix} q(0) \\ q'(0) \\ q''(0) \end{bmatrix} = \alpha T(p(x))+T(q(x))$$

so T is indeed linear. Evidently $\text{Ker}(T) = W$ by how T is defined, so W is a subspace.

Now given $\vec{x} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \mathbb{R}^3$, we can let $p(x) = a + bx + \frac{c}{2}x^2 \in \mathcal{P}_4(\mathbb{R})$ and observe that:

$$T(p(x)) = \begin{bmatrix} p(0) \\ p'(0) \\ p''(0) \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

which means T is surjective. Hence, $\text{rank}(T) = 3$. It follows that $\dim(W) = \text{nullity}(T) = 5 - 3 = 2$.

Exercise 10.7

Let $T : V \to W$ be a linear map (here V and W are allowed to be infinite-dimensional).

(a) Show that T is injective if and only if $\text{nullity}(T) = 0$.

As a result, we can also use $\text{nullity}(T)$ as a measure of how close T is to being injective: the closer $\text{nullity}(T)$ is to being 0, the closer T is to

being injective.

(b) Show that if W is finite-dimensional, then T is surjective if and only if $\text{rank}(T) = \dim(W)$.

As a result, we can also use $\text{rank}(T)$ as a measure of how close T is to being surjective (provided W is finite-dimensional): the closer $\text{rank}(T)$ is to being $\dim(W)$, the closer T is to being surjective.

(c) Find an example of vector spaces V and W and a linear map $T : V \to W$ where $\text{rank}(T) = \infty$ but T is *not* surjective.

Exercise 10.8

Let V and W be finite-dimensional vector spaces.

(a) Prove that if $\dim(V) < \dim(W)$, then it is impossible to find a surjective linear map $T : V \to W$.

(b) Prove that if $\dim(V) > \dim(W)$, then it is impossible to find an injective linear map $T : V \to W$.

(c) Prove that if $\dim(V) = \dim(W)$, then a linear map $T : V \to W$ is injective if and only if it is surjective.

Lecture 11 Isomorphisms between vector spaces

*This covers part of **Section 2.5** in the course notes.*

Fix vector spaces V and W over $\mathbb{F}$. We introduce a new concept here which is defined simply by combining injectivity and surjectivity of a linear map.

Definition 11.1 (Bijective functions and isomorphisms)

A function $f : X \to Y$ between two sets is called **bijective** when f is both injective and surjective.

A linear map $T : V \to W$ is called an **isomorphism** when T is bijective.

If a function $f : X \to Y$ is bijective, it has an inverse map $f^{-1} : Y \to X$ satisfying $f^{-1} \circ f(x) = x$ for all $x \in X$ and $f \circ f^{-1}(y) = y$ for all $y \in Y$. For an isomorphism, we can say more about the inverse.

Lemma 11.2 (Inverse of an isomorphism is an isomorphism)

If $T : V \to W$ is an isomorphism, then the inverse map $T^{-1} : W \to V$ is also an isomorphism and satisfies $T^{-1}T = \mathrm{id}_V$ and $TT^{-1} = \mathrm{id}_W$.

Proof. It is clear that we will have $T^{-1}T = \mathrm{id}_V$ and $TT^{-1} = \mathrm{id}_W$. The only thing left to check is that T^{-1} is indeed linear. Let $\vec{w}_1, \vec{w}_2 \in W$ and $\alpha \in \mathbb{F}$. Then there exist $\vec{v}_1, \vec{v}_2 \in V$ with $T\vec{v}_1 = \vec{w}_1$ and $T\vec{v}_2 = \vec{w}_2$. Now since T is bijective, we also have $T^{-1}\vec{w}_1 = \vec{v}_1$ and $T^{-1}\vec{w}_2 = \vec{v}_2$. Thus:

$$\begin{aligned} T^{-1}(\alpha\vec{w}_1 + \vec{w}_2) &= T^{-1}(\alpha T\vec{v}_1 + T\vec{v}_2) \\ &= T^{-1}T(\alpha\vec{v}_1 + \vec{v}_2) \\ &= \alpha\vec{v}_1 + \vec{v}_2 \\ &= \alpha T^{-1}\vec{w}_1 + T^{-1}\vec{w}_2 \end{aligned}$$

so T^{-1} is indeed linear. □

Definition 11.3

Let V and W be vector spaces over $\mathbb{F}$. We say V is **isomorphic** to W when there exists an isomorphism $T : V \to W$. We will write $V \cong W$ to denote that V is isomorphic to W.

An important property about the isomorphism relation $\cong$ is that it is an equivalence relation!

Proposition 11.4

$\cong$ is an equivalence relation. That is, for any vector spaces U, V, and W over $\mathbb{F}$, we have:

(i) $U \cong U$,

(ii) if $U \cong V$ then $V \cong U$, and

(iii) if $U \cong V$ and $V \cong W$, then $U \cong W$.

Proof. Exercise. □

Hence, in particular, isomorphism is a two-sided relation. Instead of saying V is *isomorphic to* W, we can just say V and W *are isomorphic*.

Example 11.5

We can show that $\mathcal{P}_3(\mathbb{F})$ and $\mathbb{F}^4$ are isomorphic. Consider the map $T : \mathcal{P}_3(\mathbb{F}) \to \mathbb{F}^4$ given by:

$$T(a + bx + cx^2 + dx^3) = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

It is easy to check that this is linear. Moreover, we see that $\text{Ker}(T) = \{0\}$, which means T is injective. By the rank-nullity theorem, we have $\text{rank}(T) = \dim(\mathcal{P}_3(\mathbb{F})) - \text{nullity}(T) = 4 - 0 = 4$, so T must also be surjective. Hence, T is an isomorphism, and this proves $\mathcal{P}_3(\mathbb{F}) \cong \mathbb{F}^4$.

Example 11.6

Consider $\mathcal{P}(\mathbb{F})$ and the subspace:

$$U = \{p(x) \in \mathcal{P}(\mathbb{F}) : p(0) = 0\}.$$

The map $T : \mathcal{P}(\mathbb{F}) \to U$ given by:

$$T(p(x)) = xp(x)$$

is easily checked to be linear and clearly indeed maps into U.

- If $T(p(x)) = 0$ then $xp(x) = 0$, which forces $p(x) = 0$ and so $\ker(T) = \{0\}$. This shows that T is injective.
- If $q(x) \in U$ then $q(0) = 0$, which means 0 is a root of $q(x)$. This must mean x is a factor of $q(x)$, and we can write $q(x) = xp(x)$ for some $p(x) \in \mathcal{P}(\mathbb{F})$. But this is precisely $T(p(x))$, so T is surjective.

Thus, T is an isomorphism.

Remark 11.7

What is isomorphism trying to convey? If V and W are isomorphic, this means not only is there is a bijective correspondence between the vectors of V and the vectors of W, but this correspondence respects the vector space operations on the spaces. Effectively this means, as far as pure vector space structure is concerned, V and W are *the same space*!

Another way to understand this is as follows: if $T : V \to W$ is an isomorphism, then W is nothing but V but with a "renaming" of vectors: every $\vec{v} \in V$ is renamed $T\vec{v}$ instead – the vector space structure still remains the same though.

In the setting of finite-dimensional spaces, we can say a lot more about how isomorphisms behave.

Proposition 11.8 (Isomorphism sends basis to basis)

Suppose V is finite-dimensional and let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for it. Let $T : V \to W$ be a linear map. Then T is an isomorphism if and only if $T\vec{v}_1, ..., T\vec{v}_n$ are distinct and $\{T\vec{v}_1, ..., T\vec{v}_n\}$ is a basis for W.

In other words, T is an isomorphism if and only if T sends a basis for V to a basis for W.

(Note that we require the $T\vec{v}_1, ..., T\vec{v}_n$ to be distinct since otherwise injectivity clearly would not hold.)

Proof. Exercise (see **Problem P4** at the end of this file). □

Corollary 11.9

If V is finite-dimensional and W is infinite-dimensional, then V is *not* isomorphic to W.

Proof. If they were isomorphic, we could find an isomorphism $T : V \to W$. If $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ is a basis, then by the preceding proposition we have that $T\vec{v}_1, ..., T\vec{v}_n$ are distinct and $\{T\vec{v}_1, ..., T\vec{v}_n\}$ is a basis for W. This means $\dim(W) = n$ as well, but this contradicts that W is infinite-dimensional. □

As it turns out, isomorphisms of finite-dimensional vector spaces are characterized *precisely* by dimension in the following way:

Theorem 11.10 ($V \cong W$ iff $\dim(V) = \dim(W)$)

Suppose V and W are finite-dimensional. Then $V \cong W$ if and only if $\dim(V) = \dim(W)$.

Proof. Suppose $\dim(V) \neq \dim(W)$. We know from **Exercise 10.8** that:

- if $\dim(V) < \dim(W)$, it is impossible to find a surjective linear map $T : V \to W$; and

- if $\dim(V) > \dim(W)$, it is impossible to find an injective linear map $T : V \to W$.

Hence, in either case, it is impossible to find an isomorphism between V and W, so $V \ncong W$.

Conversely, suppose $n = \dim(V) = \dim(W)$. Choose a basis $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ for V and a basis $\mathcal{C} = \{\vec{w}_1, ..., \vec{w}_n\}$ for W. Then we can find a (unique) linear map $T : V \to W$ satisfying $T\vec{v}_i = \vec{w}_i$. Now $\mathcal{C} = \{T\vec{v}_1, ..., T\vec{v}_n\}$ is a basis for W, and we know this means T must be an isomorphism (by **Proposition 11.8**), so we get $V \cong W$. □

From the above, it is immediate that for any vector space V and any $n \geq 0$, we have $\dim(V) = n$ if and only if $V \cong \mathbb{F}^n$ (where, by convention, $\mathbb{F}^0 = \{\vec{0}\}$, the zero vector space). But (in the case $n \geq 1$) there is more to be said about how such an isomorphism can be found!

Corollary 11.11 (Coordinate map is an isomorphism)

Let V be a finite-dimensional vector space over $\mathbb{F}$ with $\dim(V) = n \geq 1$. Let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be an ordered basis for V and let $\Phi_{\mathcal{B}} : V \to \mathbb{F}^n$ be the unique linear map which satisfies $\Phi_{\mathcal{B}}(\vec{v}_i) = \vec{e}_i$ (the ith standard basis vector of $\mathbb{F}^n$).

Then $\Phi_{\mathcal{B}}$ is an isomorphism which satisfies the following property: for all $\vec{v} \in V$, we have:

$$\Phi_{\mathcal{B}}(\vec{v}) = [\vec{v}]_{\mathcal{B}}$$

where $[\vec{v}]_{\mathcal{B}} \in \mathbb{F}^n$ denotes the coordinate vector of $\vec{v}$ relative to $\mathcal{B}$.

Proof. An application of **Proposition 11.8** tells us that $\Phi_{\mathcal{B}}$ is indeed an isomorphism. Now for any $\vec{v} \in V$, write:

$$\vec{v} = \alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n$$

for $\alpha_i \in \mathbb{F}$. Then we have:

$$\Phi_{\mathcal{B}}(\vec{v}) = \alpha_1\Phi_{\mathcal{B}}(\vec{v}_1) + \cdots + \alpha_n\Phi_{\mathcal{B}}(\vec{v}_n) = \alpha_1\vec{e}_1 + \cdots + \alpha_n\vec{e}_n = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} = [\vec{v}]_{\mathcal{B}}$$

which completes the proof. □

Exercise 11.12

Let V and W be vector spaces over $\mathbb{F}$ (which are not necessarily finite-dimensional). Let $T : V \to W$ be a linear map.

Prove that T is an isomorphism if and only if there exists a linear map $S : W \to V$ which satisfies $ST = \mathrm{id}_V$ and $TS = \mathrm{id}_W$. Show also that in this case, $S = T^{-1}$ (that is, if the inverse of T exists, it must be unique).

Exercise 11.13

(a) Consider the map $T : \mathcal{C}([0,1]) \to \mathcal{C}([0,1])$ given by $T(f)(x) = (1+x)f(x)$ for all $x \in [0,1]$.

Show that T is an isomorphism.

(b) Consider the map $S : \mathcal{C}([0,1]) \to \mathcal{C}([0,1])$ given by $S(f)(x) = xf(x)$ for all $x \in [0,1]$.

Show that S is *not* an an isomorphism.

Practice Problems

Here are some practice problems covering what we have done this week. Problems **P6** and **P7** are just for fun – you will not need to know the content these questions cover for any assessments.

P1 Consider the space

$$\mathbb{F}^\infty = \{(a_n)_{n=1}^\infty : \text{ each } a_n \in \mathbb{F}\}$$

of sequences in $\mathbb{F}$ with the usual addition and scalar multiplication. Define:

- the **left shift operator** $L : \mathbb{F}^\infty \to \mathbb{F}^\infty$ by $L((a_1, a_2, a_3, ...)) = (a_2, a_3, a_4, ...)$;
- the **right shift operator** $R : \mathbb{F}^\infty \to \mathbb{F}^\infty$ by $R((a_1, a_2, a_3, ...)) = (0, a_1, a_2, a_3, ...)$.

(a) Verify that L and R are both linear operators on $\mathbb{F}^\infty$.

(b) Determine $\text{Ker}(L)$ and $\text{Ker}(R)$. Are either of L or R injective?

(c) Determine $\text{Range}(L)$ and $\text{Range}(R)$. Are either of L or R surjective?

P2

(a) Show that it is impossible to find a linear operator $T : \mathbb{R}^{69} \to \mathbb{R}^{69}$ satisfying $\text{Ker}(T) = \text{Range}(T)$.

(b) Find an example of a linear operator $T : \mathbb{R}^4 \to \mathbb{R}^4$ satisfying $\text{Ker}(T) = \text{Range}(T)$.

P3 Let $T : \mathcal{M}_{2\times 4}(\mathbb{F}) \to \mathcal{P}_4(\mathbb{F})$ be a linear map and suppose U is a subspace of $\mathcal{M}_{2\times 4}(\mathbb{F})$ with $\dim(U) = 3$. Show that T must be surjective.

P4 Let V and W be vector spaces over $\mathbb{F}$ with $\dim(V) = n$ and let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for V. Let $T : V \to W$ be a linear map.

(a) Prove that T is injective if and only if $T\vec{v}_1, ..., T\vec{v}_n$ are distinct and $\{T\vec{v}_1, ..., T\vec{v}_n\}$ is a linearly independent set in W.

(b) Prove that T is surjective if and only if $\{T\vec{v}_1, ..., T\vec{v}_n\}$ is a spanning set for W.

(c) Explain how **Proposition 11.8** follows from **(a)** and **(b)**.

P5 Let V be a vector space with $n = \dim(V)$ and let $T : V \to V$ be a linear operator.

(a) Show that if $T^2 = TT = O$, then $\text{rank}(T) \leq \dfrac{n}{2}$.

(Hint: if $T^2 = O$, what does this mean about $\text{Ker}(T)$ and $\text{Range}(T)$?)

(b) **(challenging!)** Show that if $T^3 = O$, then $\text{rank}(T) \leq \dfrac{2n}{3}$.

P6 This problem is an extension of **Exercise 9.14**, in which we proved that the map $T : \mathcal{C}^1([0,1]) \to \mathcal{C}([0,1])$ given by

$$T(f) = f + f'$$

is linear and has $\text{Ker}(T)$ of dimension 1 (in other words, $\text{nullity}(T) = 1$).

Now let:

$$\mathcal{C}^2([0,1]) = \{f \in \mathcal{C}^1([0,1]) : f' \text{ is differentiable and } f'' \text{ is continuous}\}.$$

where f'' is the second derivative of f. It can be shown that $\mathcal{C}^2([0,1])$ is a subspace of $\mathcal{C}^1([0,1])$. Define $S : \mathcal{C}^2([0,1]) \to \mathcal{C}([0,1])$ by:

$$S(f) = f + f''.$$

(a) Show that S is linear.

(b) Show that $\cos x$ and $\sin x$ are both in $\text{Ker}(S)$ and that they are linearly independent.

(c) Show that if $h \in \text{Ker}(S)$ and satisfies $h(0) = h'(0) = 0$, then $h = 0$.

(Hint: consider $H(x) = (h(x))^2 + (h'(x))^2$ and show that $H' = 0$. What does this tell you about H?)

(d) Let $g \in \text{Ker}(S)$ and let $a = g(0)$ and $b = g'(0)$. Show that $g(x) = a\cos x + b\sin x$. Conclude therefore that $\text{Ker}(S) = \text{Span}\{\cos x, \sin x\}$ and $\text{nullity}(S) = 2$.

(Hint: apply part **(c)** to $h(x) = g(x) - a\cos x - b\sin x$.)

(e) Between T and S, which would you say is "more injective"?

P7 Recall that for vector spaces V and W, the set:

$$\mathcal{L}(V, W) = \{T : V \to W \mid T \text{ is linear}\}$$

is a vector space under the operations of addition and scalar multiplication on linear maps (see **Proposition 8.10** in **Week 3**).

We consider here the special case where V has finite dimension n and $W = \mathbb{F}$. We define:

$$V^* = \mathcal{L}(V, \mathbb{F})$$

and call V^* the **dual space** of V. The elements of V^* are therefore *linear maps* of the form $\varphi : V \to \mathbb{F}$ (that is, linear maps sending vectors in V to scalars in $\mathbb{F}$).

Let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for V.

(a) Show that there exist unique vectors $\varphi_1, ..., \varphi_n \in V^*$ which satisfy:

$$\varphi_i(\vec{v}_j) = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

for every $1 \leq i, j \leq n$.

(b) Show that $\mathcal{B}^* = \{\varphi_1, ..., \varphi_n\}$ as defined above forms a basis for V^*. We call $\mathcal{B}^*$ the **dual basis** of $\mathcal{B}$. Conclude that V is isomorphic to V^*.

(c) This is where things get pretty weird!

Let $V^{**} = (V^*)^*$, the dual space of the dual space of V. V^{**} is therefore the space of linear maps which send linear maps from V to $\mathbb{F}$ into $\mathbb{F}$! We call V^{**} the **double dual** of V.

By part **(b)**, we have $\dim(V^{**}) = \dim(V^*) = \dim(V) = n$, so $V \cong V^{**}$. But we can say more:

For each $\vec{v} \in V$, let $\text{ev}_{\vec{v}} : V^* \to \mathbb{F}$ be given by:

$$\text{ev}_{\vec{v}}(\varphi) = \varphi(\vec{v})$$

for every $\varphi \in V^*$. Show that $\text{ev}_{\vec{v}}$ is linear and therefore an element of V^{**}.

(d) Show that the map $\Phi : V \to V^{**}$ given by $\Phi(\vec{v}) = \text{ev}_{\vec{v}}$ is an isomorphism!

(The interesting part here is that we didn't need to choose a basis for V to find this isomorphism. On the other hand, constructing an isomorphism from V to V^* in general requires a choice of basis for V.)